Social Network Analysis

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Reading

Required reading:

- Aggarwal Chapter 19 (relevant topics described in these slides)

Acknowledgement:

Unless stated otherwise, the contents and images on the following slides are sourced from

C. C. Aggarwal, Data Mining: The Textbook. Springer, 2016.

What is social network?

- Online networks such as Twitter, LinkedIn, and Facebook

Indirect forms of social networks:

- Telecommunications, email, and electronic, chat messengers.
- Sites that are used for sharing online media content, such as Flickr, YouTube, or Delicious
- Scientific communities are organized into bibliographic and citation networks.

Ways to analyze networks

- **Identify densely linked clusters of nodes**
  - Community detection (clustering)
- **Predict the type/color of a given node**
  - Collective classification (Node classification)
- **Predict whether two nodes are linked**
  - Link prediction (recommender)
- **Measure similarity of two nodes/subgraphs**
  - subgraph similarity

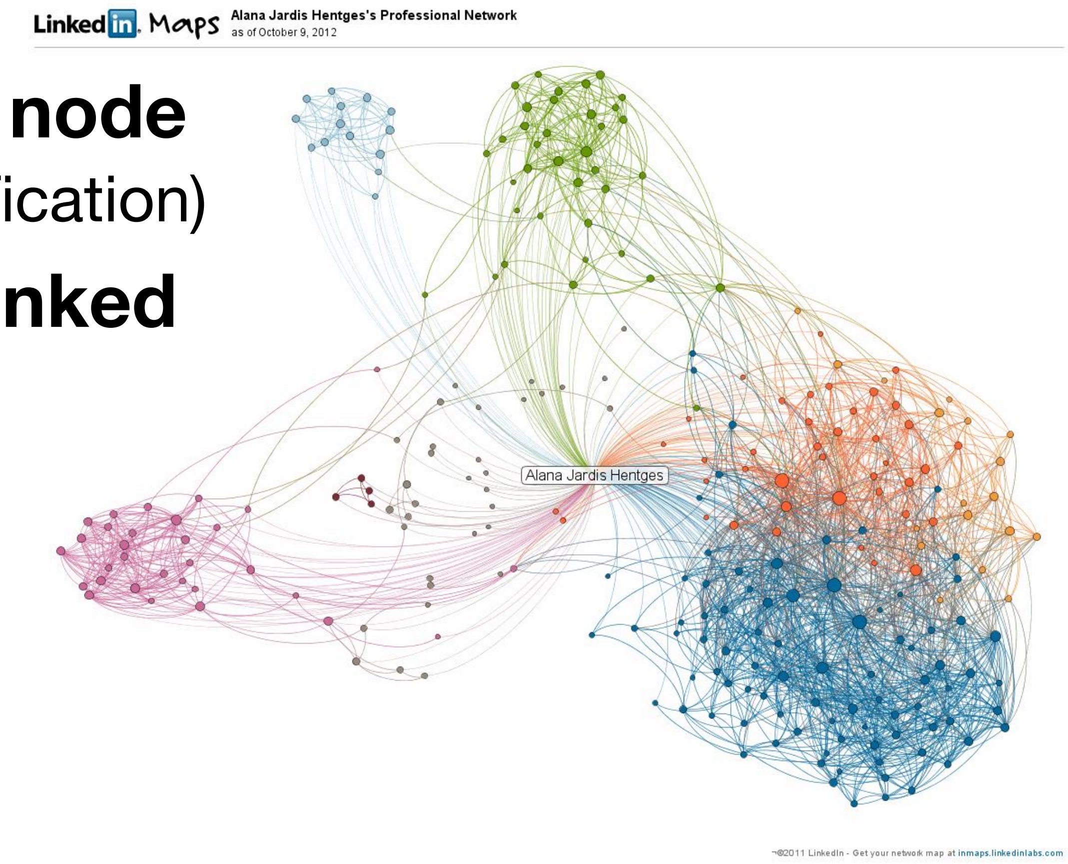


Image source: https://blogs.pugetsound.edu/cesblogs/files/2012/10/Alana-Linked-In-Map.gif

Social Networks

A social network can be structured as a graph $G = (N,A)$, where N is the set of nodes and A is the set of edges. Each individual in the social network is represented by a node in N, and is also referred to as an *actor*. The edges represent the connections between the different actors.

Properties:

- Homophily: nodes that are connected to one another are more likely to have similar properties. "*Birds of a feather flock together*"
- The principle of triadic closure: *If two individuals in a social network have a friend in common, then it is more likely that they are either connected or will eventually become connected in the future.*
- The local clustering coefficient $\eta(i)$ of node i in a graph is the fraction of all possible node-pairs that have an edge between the nodes connected to node i. The Watts–Strogatz *network average clustering coefficient* is the average value of $\eta(i)$ over all nodes in the network. The triadic closure property increases the clustering coefficient

Properties (continue)

Dynamics of Network Formation

- *Preferential attachment:* In a growing network, the likelihood of a node receiving new edges increases with its degree.
- *Small world property:* the average path length between any pair of nodes is quite small. Typically, for a network containing $n(t)$ nodes at time t, many models postulate that the average path lengths grow as $\log(n(t))$.

See en.wikipedia.org/wiki/Six_degrees_of_separation

- *Densification:* Almost all real-world networks such as the Web and social networks add more nodes and edges over time than are deleted. If $n(t)$ is the number of nodes in the network at time t, and $e(t)$ is the number of edges, then the network exhibits the following *densification power law*: $e(t) \propto n(t)^{\beta}$, where the exponent β is a value between 1 and 2.
- *Shrinking diameters:* In most real-world networks, as the network densifies, the average distances between the nodes shrink over time.
- *Giant connected component:* As the network densifies over time, a giant connected component emerges.

Properties (continue)

- **Power-Law Degree Distribution:** the proportion of nodes $P(k)$ with degree k, is regulated by the following *power-law degree distribution*: $P(k) \propto k^{-\gamma}$, where the value of parameter γ ranges between 2 and 3.
- **Measures of Centrality and Prestige:**
- Nodes that are central to the network have a significant impact on the properties of the network, such as its density, pairwise shortest path distances, connectivity, and clustering behavior. Many of these nodes are hub nodes, with high degrees that are a natural result of the dynamical processes of large network generation.
- A related notion of centrality is *prestige.* For example, on Twitter, an actor with a larger number of followers has greater prestige. On the other hand, following a large number of individuals does not bring any prestige but is indicative of the *gregariousness* of an actor.

Example Power-Law Degree Distribution

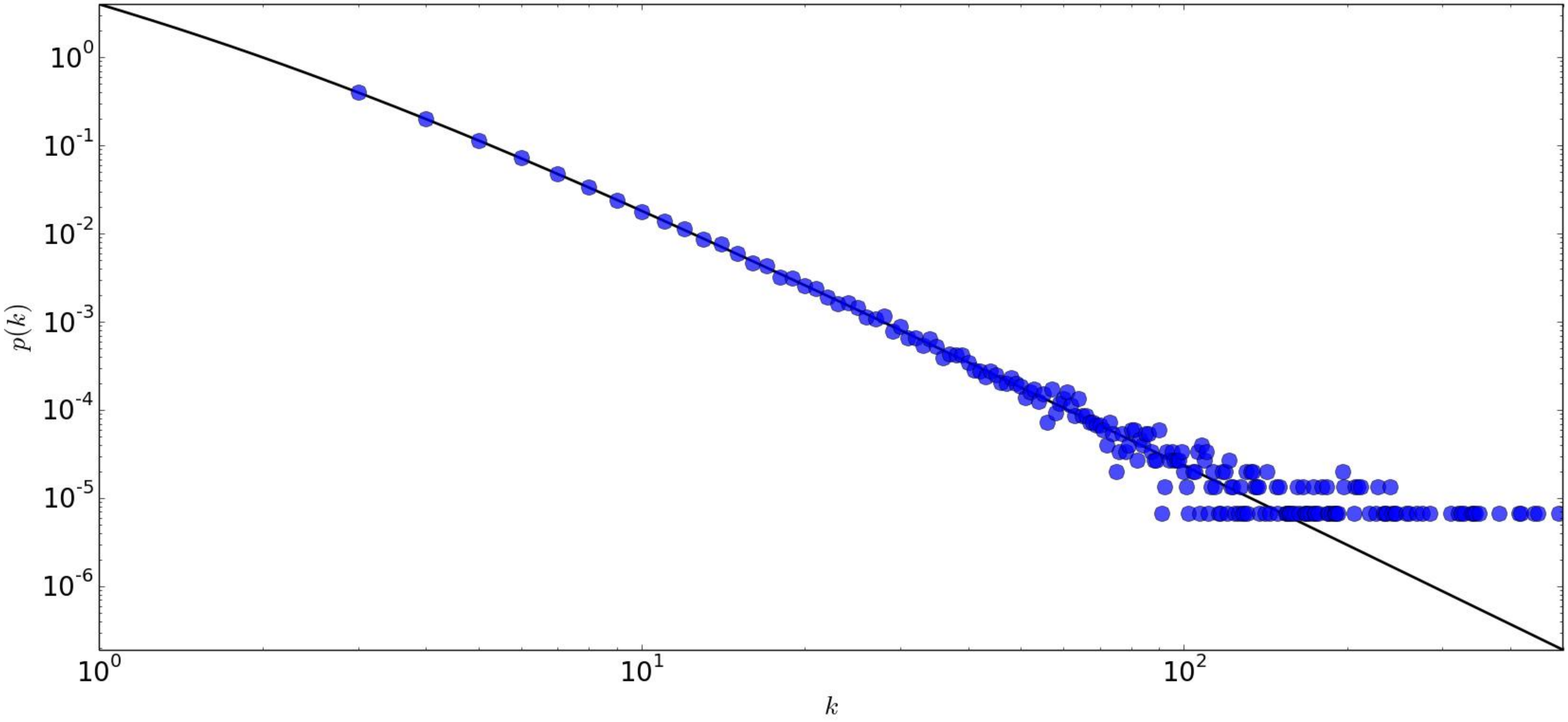


Image source: en.wikipedia.org/wiki/Scale-free_network

Community Detection

- An approximate synonym for “clustering” in the context of social network analysis.
- Community detection is one of the most fundamental problems in social network analysis.
- In the social network domain, network clustering algorithms often have difficulty in cleanly separating out different clusters because of some natural properties of typical social networks (see next slide)

Factors of difficulties in clustering of networks

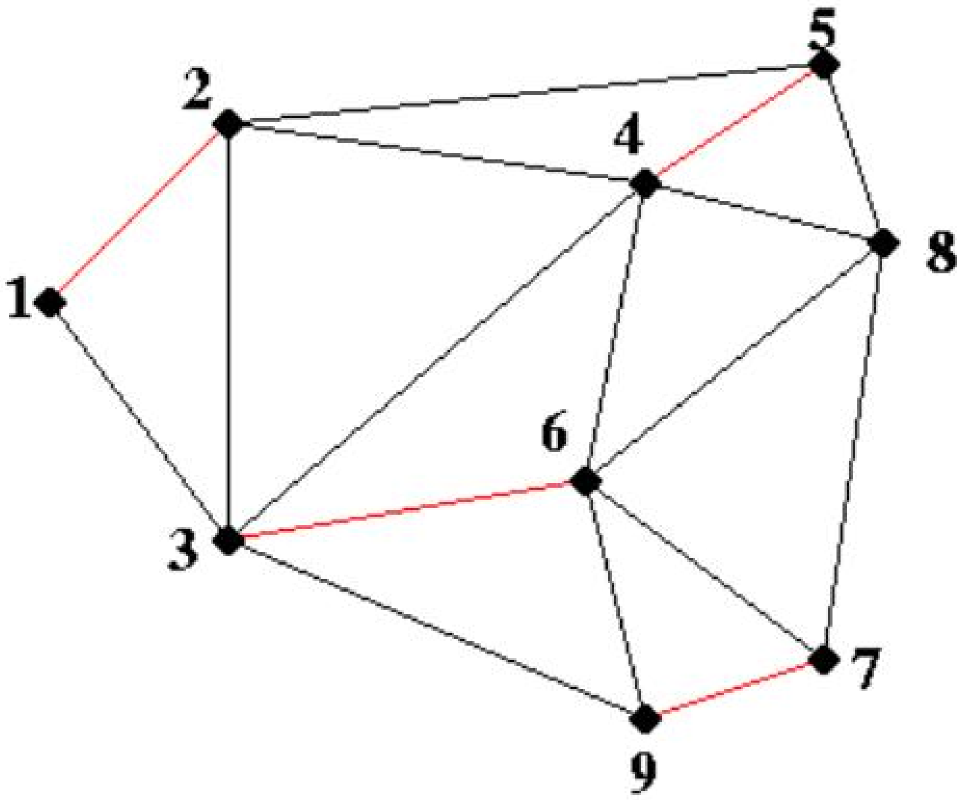
1. The distance-based *k*-means algorithm cannot be easily generalized to networks.
2. Hubs complicate the clustering process
3. Different parts of the social network have different edge densities.
4. Real social networks often have a giant component that is densely connected. This contributes further to the tendency of community detection algorithms to create imbalance clusters.

Network clustering or community detection

- The basic objective is to partition a given network into *k* sets of nodes, such that the sum of the weights of the edges with end points in different partitions is minimized.
- Some variations, e.g., *partition balancing* in which different clusters have similar numbers of nodes.
- In the special case, where every edge weight is 1, and there are no balancing constraints on partitions, the 2-way cut problem is polynomially solvable.
- Incorporating arbitrary edge weights or balancing constraints makes the problem NP-hard. Many network clustering algorithms focus on balanced 2-way partitioning of graphs.
- A 2-way partitioning can be recursively used to generate a *k*-way partitioning.

The 2-way cut problem

What is the best 2-way minimum cut for this graph?



Maximal matching given by red edges:

Any additional edge will connect to one of the nodes already present

Categories of network clustering

Two broad categories of methods:

1. Local methods: the Kernighan–Lin algorithm, and Fiduccia-Mattheyses algorithms, which were the first effective 2-way cuts by local search strategies. Key drawback: random initialization.
2. Global methods rely on global properties of the entire graph and don't rely on random initialization. The most common example is spectral partitioning, where a partition is derived from approximate eigenvectors of the adjacency matrix, or **spectral clustering** that groups graph vertices using the eigendecomposition of the graph Laplacian matrix.

A multi-level graph partitioning algorithm works by applying one or more stages. Each stage reduces the size of the graph by collapsing vertices and edges, partitions the smaller graph, then maps back and refines this partition of the original graph.

Source: https://en.wikipedia.org/wiki/Graph_partition

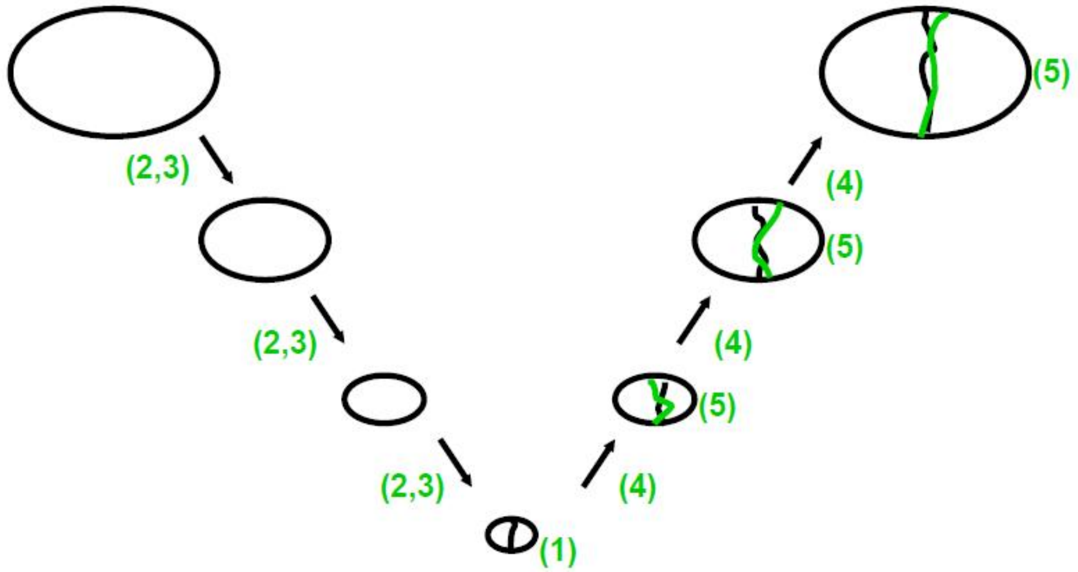
Algorithm $KernighanLin$(Graph: $G = (N, A)$, Weights:$[w_{ij}]$)
begin
Create random initial partition of N into N_1 and N_2;
repeat
Recompute D_i values for each node $i \in N$;
Unmark all nodes in N;
for $i = 1$ to $n/2$ **do**
begin
Select $x_i \in N_1$ and $y_i \in N_2$ to be the unmarked node pair with the highest exchange-gain $g(i) = J_{x_i y_i}$;
Mark x_i and y_i;
Recompute D_j for each node j, under the assumption that x_i and y_i will be eventually exchanged;
end
Determine k that maximizes $G_k = \sum_{i=1}^{k} g(i)$;
if $(G_k > 0)$ **then** exchange $\{x_1 \dots x_k\}$ and $\{y_1 \dots y_k\}$ between N_1 and N_2;
until $(G_k \leq 0)$;
return(N_1, N_2);
end

The Gain by moving a node i from one partition to another:
$D_i = E_i$ (external cost) – I_i (internal cost)

Multilevel Partitioning - High Level Algorithm

(N+,N-) = Multilevel_Partition(N, E)
    ... recursive partitioning routine returns N+ and N- where N = N+ U N-
    if |N| is small
(1)   Partition G = (N,E) directly to get N = N+ U N-
        Return (N+, N-)
    else
(2)   Coarsen G to get an approximation $G_C = (N_C, E_C)$
(3) (N_C+ , N_C-) = Multilevel_Partition(N_C, E_C)
(4)   Expand (N_C+ , N_C-) to a partition (N+ , N-) of N
(5)   Improve the partition (N+ , N-)
        Return (N+ , N-)
    endif

"V - cycle:"



How do we
- Coarsen?
- Expand?
- Improve?

Image attributed to Juan J. Alonso.

Multilevel Kernighan-Lin

- Coarsen graph and expand partition using maximal matchings
- Improve partition using Kernighan-Lin

This is the algorithm that is implemented in Metis

- *Definition*: A matching of a graph G(N,E) is a subset E_m of E such that no two edges in E_m share an endpoint
- *Definition:* A maximal matching of a graph G(N,E) is a matching E_m to which no more edges can be added and remain a matching

Example of maximum matching

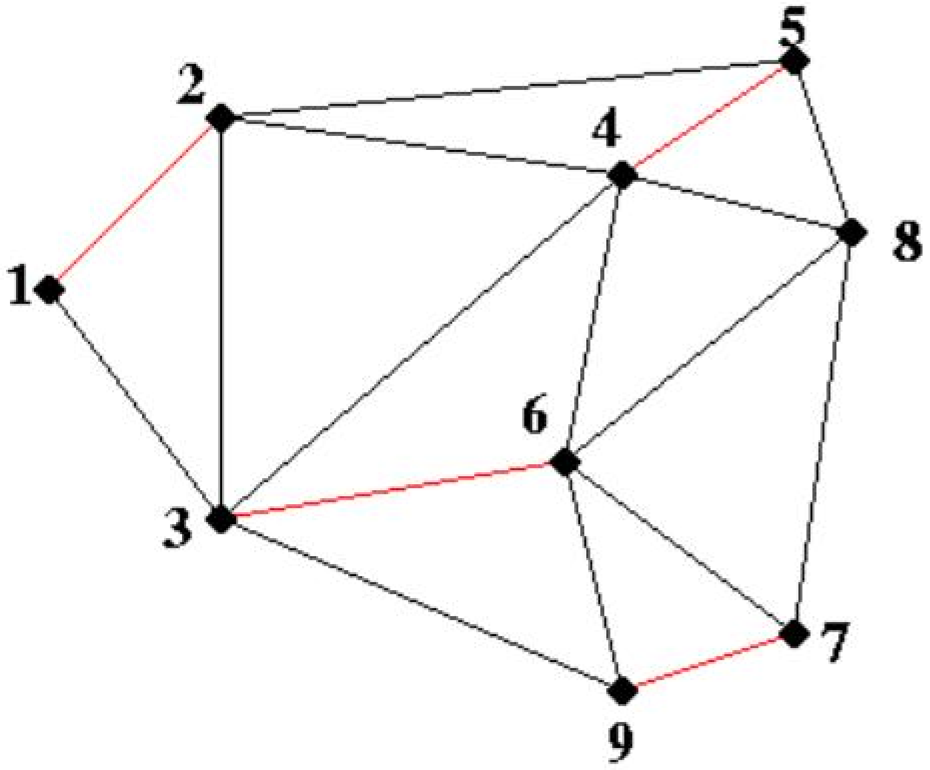


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Image attributed to Juan J. Alonso.

Example of Coarsening using maximum matching

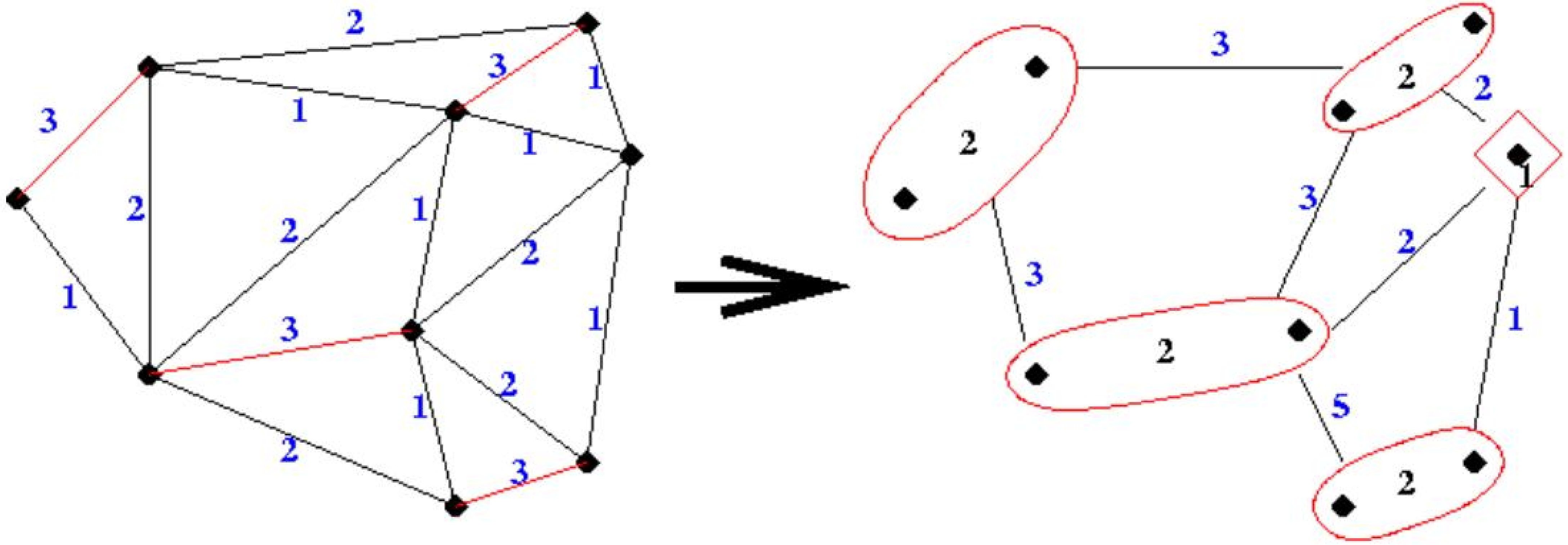


Image attributed to Juan J. Alonso.

Different ways to coarsen a graph

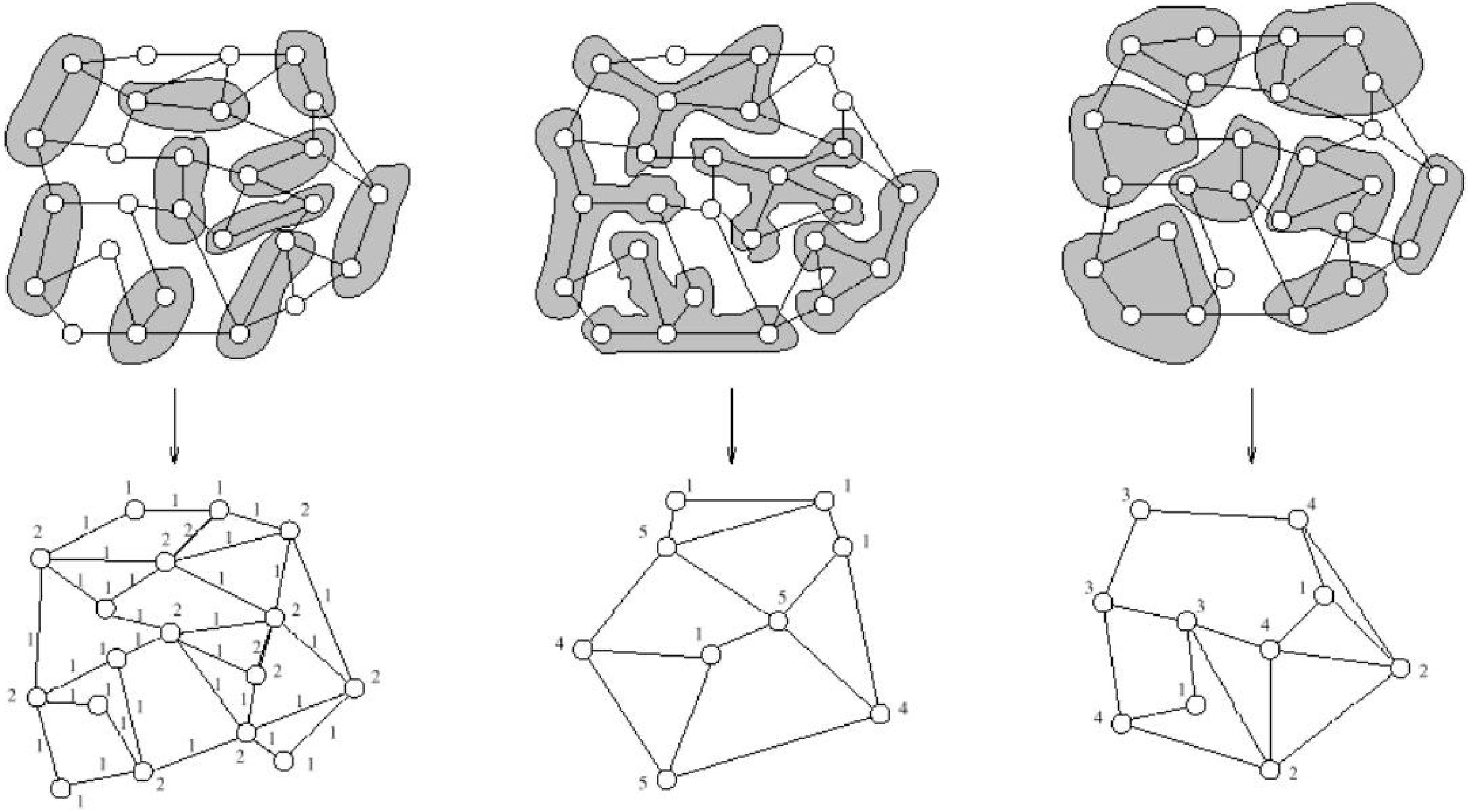


Image attributed to Juan J. Alonso.

Spectral Clustering

- The spectral method uses a graph embedding approach, so that the local clustering structure of the network is preserved by the embedding of the nodes into multidimensional space. The idea is to create a multidimensional representation of the graph so that a standard *k*-means algorithm can be used on the transformed representation.

Spectral Clustering (three basic steps)

1. Preprocessing
   - Produce a similarity matrix of a neighborhood graph

2. Eigen-decomposition
   - Compute eigenvalues & eigenvectors of the similarity matrix
     or Laplacian matrix
   - Map each point in the input space to the new space,
     represented by k eigenvectors

3. Clustering (often uses **k**-means)

Laplacian matrix & Objective function

- *Laplacian matrix L* of weight matrix W: $L = \Lambda - W$, where Λ is a diagonal matrix satisfying $\Lambda_{ii} = \sum_{j=1}^{n} w_{ij}$ which is the sum of the weights of the edges incident on node *i*, can be viewed as the local density of the network at node *i* (or node-degree matrix Λ).
- Optimization objective:

$$\textbf{Minimize trace}(\mathbf{Y}^{\mathbf{T}}\mathbf{L}\mathbf{Y})$$

$$\textbf{subject to } \mathbf{Y}^{\mathbf{T}}\mathbf{\Lambda}\mathbf{Y} = \mathbf{I}$$

Y is an $n \times k$ matrix of *k*-dimensional embedding.

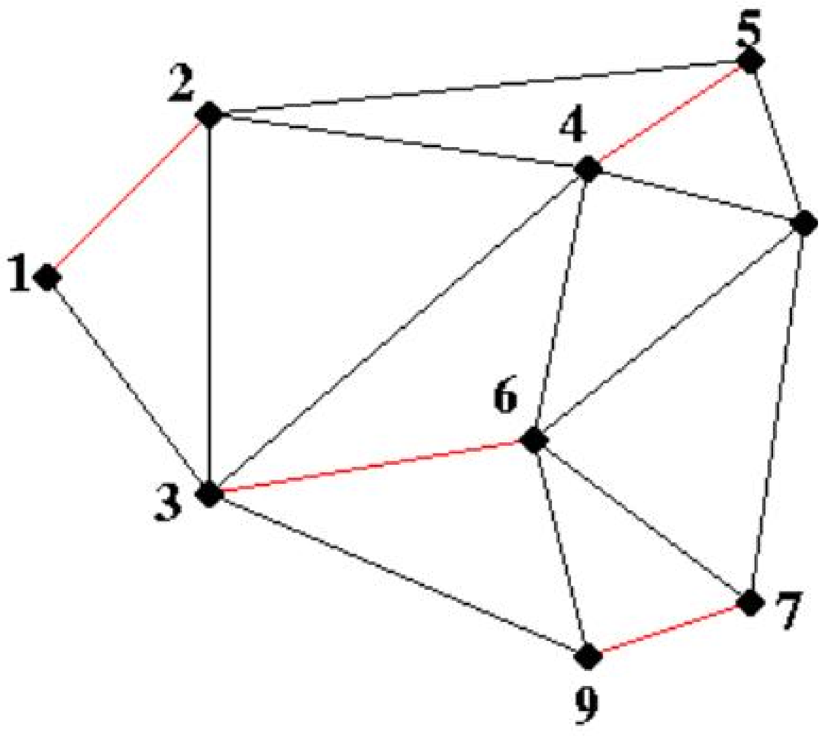
The objective effectively minimizes the distances between highly connected nodes.

Interpretations

- The optimal solutions for these k column vectors can be shown to be proportional to the successive directions corresponding to the (not necessarily orthogonal) right eigenvectors of the asymmetric matrix $\Lambda^{-1}L$ with increasing eigenvalues.
- After discarding the first trivial eigenvector e_1 with eigenvalue $\lambda_1 = 0$, this results in a set of k eigenvectors $e_2, e_3 \ldots e_{k+1}$, with corresponding eigenvalues $\lambda_2 \leq \lambda_3 \ldots \leq \lambda_{k+1}$.
- A simpler way of understanding normalization is by examining the similarity matrix $\Lambda^{-1/2}W\Lambda^{-1/2}$ whose large eigenvectors yield the normalized spectral embedding. $[\, L = I - \Lambda^{-1/2}W\Lambda^{-1/2} \,]$
- In this matrix, the edge similarities are normalized by the geometric mean of the node degrees at their end points. This can be viewed as a normalization of the edge similarities with a local measure of the network density.
- Normalization will yield more balanced clusters with widely varying density over the network. It also reduces the bias towards high degree nodes.

Discussion #1 (Clustering)

1. Explain what a 2-way minimum cut problem is.
2. *Why is the balancing constraint more important in community detection algorithms, as compared to multidimensional clustering algorithms? What would the result of an unconstrained minimum 2-way cut look like in a typical real network?*
3. Explain why the Kernighan-Lin algorithm is a local method. Does it produce balanced clusters?
4. Explain why the Spectral Clustering is a global method. Does it produce balanced clusters?
5. The last step of the Spectral Clustering procedure often employs k-means clustering. Explain why this is the clustering algorithm of choice (and not other clustering algorithms).
6. Must we use k-means as the final step in the Spectral Clustering procedure? Explain the impact (if any) of using a different clustering algorithm.



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  until $(G_k \leq 0)$;
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end

Discussion #2 (Clustering)

7. Describe the differences in using graph kernels to deal with a large graph versus a database of graphs.
8. Describe the differences in using spectral clustering to deal with a large graph versus a database of graphs.
9. One of the drawback of Kernighan-Lin is: random initialization. Does spectral clustering has the same drawback?
10. The term “spectral clustering” is a misnomer because the “spectral” component performs feature transformation only and a separate clustering algorithm must be used to perform the actual clustering. Discuss this statement (whether it is true or false, and explain your reasoning)

Collective Classification (Node classification)

- In a network where only some nodes are labelled with a class label, the task is to predict an unlabeled node.
- The solution to this model is crucially dependent on the notion of homophily.
- An issue to be resolved is the sparsity of node labels: One must not only use the direct connections to labeled nodes, but also use the indirect connections through unlabeled nodes.
- Collective classification in networks are always performed in a *transductive semisupervised setting*, where the test instances and training instances are classified jointly.
- *It is important not only from the perspective of social network analysis, but also for semisupervised classification of any data type.*

Three distinct types of correlations

1. The correlations between the label of node *v* and the features of *v*.
2. The correlations between the label of *v* and the features (including observed labels) of nodes in the neighborhood of *v*.
3. The correlations between the label of *v* and the unobserved labels of nodes in the neighborhood of *v*.

Collective classification refers to the combined classification of a set of interlinked objects using these three types of information.

Iterative Classification Algorithm (ICA)

- An important step of the *ICA* algorithm is to derive a set of *link features* in addition to the available content features in X_i. The most important link features correspond to the distribution of the classes in the immediate neighborhood of the node.
- For each node i, its adjacent node j is weighted by w_{ij} for computing its credit to the relevant class.
- It is also possible, in principle, to derive other link features based on structural properties of the graph such as the degree of the node, *PageRank* values, or connectivity features.

Algorithm ICA(Graph $G = (N, A)$, Weights: $[w_{ij}]$, Node Class Labels: $\mathcal{C}$,
Base Classifier: $\mathcal{A}$, Number of Iterations: T)

begin

repeat

Extract link features at each node with current training data;

Train classifier $\mathcal{A}$ using both link and content features of current training data and predict labels of test nodes;

Make (predicted) labels of most "certain" n_t/T test nodes final, and add these nodes to training data, while removing them from test data;

until T iterations;

end

Supervised Spectral Methods

Two different ways for collective classification of graphs

- The first method directly transforms the graph to multidimensional data to facilitate the use of a multidimensional classifier such as a k-nearest neighbor classifier.
- The second method directly learns an $n \times k$ class probability matrix Z with an optimization formulation related to spectral clustering.

Supervised Feature Generation with Spectral Embedding

Let $G = (N, A)$ be the undirected graph with weight matrix W. The approach consists of the following steps, the first of which is to augment G with class-based supervision:

1. Add an edge with weight μ between each pair of nodes with the same label in G. If an edge already exists between a pair of such nodes, then the two edges are consolidated by adding μ to the weight of the existing edge. The resulting graph is denoted by $G+$. The parameter μ controls the level of supervision from existing labels.
2. Use the spectral embedding approach to generate an r-dimensional embedding of the augmented graph $G+$.
3. Apply any multidimensional classifier, such as a nearest neighbor classifier, on the embedded data.

Discussion #3 (Classification)

1. Describe the distinctive features of ICA in comparison with other collective classification methods.
2. Can we use graph kernels for collective classification? How can we do it?
3. How does the graph kernel approach differ from the spectral embedding based approach for collective classification?

Reflection

- What have you learned this week?
  (Every student shall attempt to answer this question.)

- What have you learned about the treatments for a single network versus a dataset of small graphs?